COURSE 2 · SECTION 3 · WORKED SOLUTION

Where does the NPD schedule slip?

Section 3 · New product development: concept to release

DESIGN QUESTION

Across all 16 part revisions, which NPD stage misses its planned end most often, where are the slip days, and how much of the P300-B drawing delay came from Checking?

Steps

1. Open Section04_Start.pbix (or Engxtech_S03.pbix) > File > Save as > Engxtech_S03_Exercise.
2. Data pane > NPD_Stages: the columns and measures below are already there from the video. Right-click NPD_Stages > New measure: Rework Gates.
3. Table visual: Stage, On-time Gate %, Total Slip Days, Median Plan Days, Median Actual Days. Sort by Stage.
4. NPD page: PartRev slicer = P300-B. The Gantt is a stacked bar: Y axis Stage; X axis Actual Offset (transparent) then Actual Days.
5. Card with Rework Gates (or a table with GateResult and Count of PartRev).

DAX calculated columns (NPD_Stages, from the video)

```
Plan Days   = DATEDIFF ( NPD_Stages[PlannedStart], NPD_Stages[PlannedEnd], DAY )
Actual Days = DATEDIFF ( NPD_Stages[ActualStart], NPD_Stages[ActualEnd], DAY )
Slip Days   = MAX ( 0, NPD_Stages[Actual Days] - NPD_Stages[Plan Days] )
On Time = IF ( NOT ISBLANK ( NPD_Stages[ActualEnd] ),
               IF ( NPD_Stages[ActualEnd] <= NPD_Stages[PlannedEnd], 1, 0 ) )
Rev Start =
VAR pr = NPD_Stages[PartRev]
RETURN MINX ( FILTER ( NPD_Stages, NPD_Stages[PartRev] = pr ), NPD_Stages[PlannedStart] )
Actual Offset = DATEDIFF ( NPD_Stages[Rev Start], NPD_Stages[ActualStart], DAY )
```

DAX measures (NPD_Stages)

```
On-time Gate % = DIVIDE ( SUM ( NPD_Stages[On Time] ), COUNT ( NPD_Stages[On Time] ) )   -- video
Total Slip Days = SUM ( NPD_Stages[Slip Days] )                                          -- video
Rework Gates = CALCULATE ( COUNTROWS ( NPD_Stages ), NPD_Stages[GateResult] = "Rework" )  -- new
```

COUNT ignores blanks, so On-time Gate % only counts rows with an ActualEnd (all 192 at the data cut-off). Checking and Design review are where gates fail: for P300-B the drawing went round review 3 times, so both gates show Rework (2 loops). Do not sum ReworkLoops over both stages; each loop repeats both, so count one stage only.

Expected result

Check	Result
On-time gates, all stages	**56.3 % (108/192)**
On-time gates, Checking	**37.5 % (6/16)**
Stage with the most slip days	**Checking 31 d**
Slip days, all stages	**110 d**
Checking median planned / actual days	**4.0 / 6.0**
P300-B slip in Checking	**10 d**
P300-B slip, stages 1-10 (= drawing lateness)	**19 d**
Rework gates	**16**